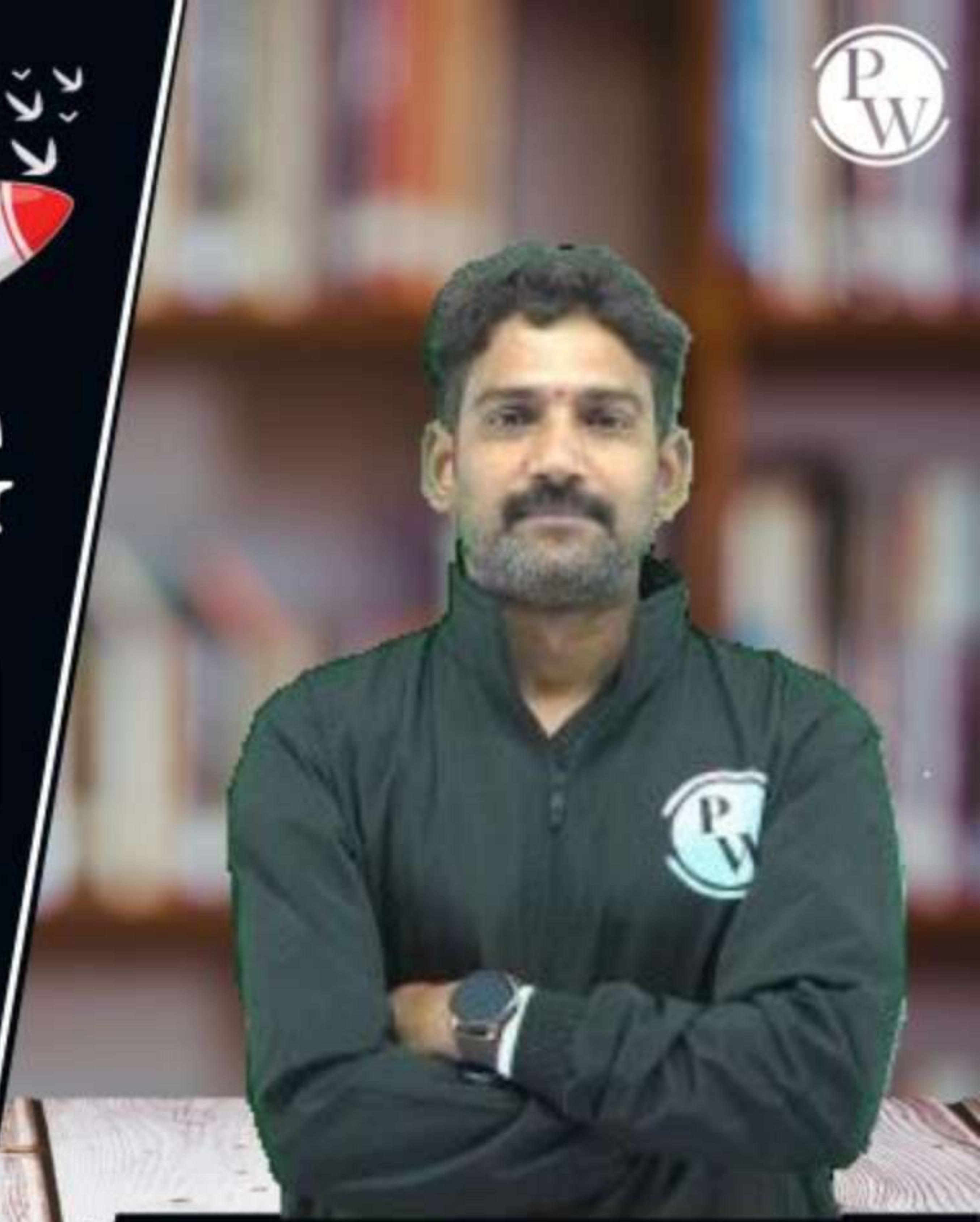
CS & IT ENGINEERING

C Programming

Practice Session

Lecture No.- 01



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Recap of Previous Lecture

- Switch-case
- Iterative control stmts
  - do-while
  - while
  - for

Topics to be Covered

- Examples on Loops
- Unconditional Control Stmts
- Practice Session

Examples

```
int main( )
{
    int i, j, sum=0;
    for(i=1; i<5; i++)
    {   sum= sum+i;
        j=1;
        while( j<5 )
        {
            sum= sum+j;
            j= j+i;
        }
    }
    Printf("%d", sum);
} return 0;
```

o/p = 30

i	j	Sum
~~1~~ ~~2~~ ~~3~~ ~~4~~ 5	~~1~~ ~~2~~ ~~3~~ ~~4~~ ~~1~~ ~~3~~ ~~5~~ ~~1~~ ~~4~~ ~~7~~ ~~1~~ 5	~~0~~ ~~1~~ ~~2~~ ~~4~~ ~~7~~ ~~11~~ ~~13~~ ~~14~~ ~~17~~ ~~20~~ ~~21~~ ~~25~~ ~~29~~ 30

i=1
sum=0+1=1
j=1, 1<5 True
sum=1+1=2
j=2
sum=2+2=4
sum= 4+3+4
=11

i=2
sum=13
j=1 sum=14
j=1+2=3
sum=14+3=17
j=3+2=5

i=3
sum=17+3=20
j=1, sum=21
j=1+3=4
sum=21+4
=25
j=4+3=7

i=4
sum=25+4=29
j=1, sum=30
j=1+4=5
5<5 false

i=5
5<5 False

```
void main( )
{
    int a, b, c;

    for(a=1; a<=5; a++);        ⇒ for( ; ; ) { ; }

    for (b=1; b<=10; b++);      for ( ; ; ) { ; }

        c= a+b;     ⇒ 6+11=17

    printf("%d", c);
}
```

a=6

b=11

o/p = 17

ignored

Decision making

```
void main( )
{
    int i, j, result=1;

    for( i=1, j=7 ; i<=5, j>=1; i++, j--)
        result = result + j-i;

    printf("%d", result);
}
```

o/p = 1

j=7, i=1

result= 1+7-1 = 7

i=2, j=6

result= 7+6-2= 11

i=3, j=5

result= 11+5-3= 13

i=4 j=4

result = 13+4-4 =13

i=5 j=3

result = 13+3-5 = 11

i=6, j=2

result= 11+2-6 = 7

i=7, j=1

result= 7+1-7

=1

j=0

0 >=1 False

Topic : Control Statements - 3

```
int main( )
{
  int i, count=0;

  for(i=0; i<5; i++)
  {
     switch(i)
     {
        case 1 :  count ++;

        case 0 :  count += 2;

        case 2 :  count += i;  break;

        case 4 :  count --;

        default :  count -= i;
     }
  } i++;
  printf("%d", count);
  return 0;
}
```

o/p = -1

$i=0$ count = 2
count = 2+0 = 2
i++ ⇒ i = 1

$i=2$ count = 2+2 = 4
i = 3

$i=4$ count = 3
count = 3-4 = -1
i = 5

$i=6$

Unconditional control statements / Jumping statements

- break, goto, Continue, return

break :- It moves control to out of block, where break is Executed.
- Unidirectional control statement (↓)

Continue :- It causes control to go to the beginning of next iteration.
- It can be only used in Loops.
- It is also an unidirectional Control statement (↑)

goto :- It causes the control to go to the given Label.
- It is bi-directional Control statement (↕)

return :- It is used to return value (or) control to the Caller function.
- It is also bidirectional ctrl stmt. (↕)

Examples

```
int main( )
{
  int i;
  for(i=1; i<10; i++)
  {
     if((i==3)||(i==7))
        break;
     Printf("%d", i);
  }
  return 0;
}
```

o/p : 1 2

```
int main( )
{
  int i;
  for( i=1; i<10; i++)
  {
     if((i==3)||(i==7))
        Continue;
     Printf(" %d", i);
  }
  return 0;
}
```

o/p : 1 2 4 5 6 8 9

```
int main( )
{
  int i=1, n=3;
xyz:
  Pf("%d* %d= %d", n, i, (n*i));
  i++;
  if( i<=10)
    goto xyz;
  return 0;
}
```

o/p :
3*1=3
3*2=6
3*3=9
3*4=12
3*5=15
3*6=18
3*7=21
3*8=24
3*9=27
3*10=30

$x=18,\ y=22\quad GCD(x,y)=2$

#Q. Consider the following C program

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Let $x=21,\ y=14$

```
main() {
    int x, y, m, n;
    scanf ("%d %d", &x, &y);
    /* Assume x > 0 and y > 0 */
    m = x;
    n = y;
    while (m! = n)  {
      if (m > n)
        m = m - n;
      else
        n = n - m;  }
    print f ("% d", n);
}
```

$m = \cancel{21}\ 7$ $\quad n = \cancel{14}\ 7$

$21 \neq 14$ True | $7 \neq 14$ True | $7 \neq 7$ False

$21 > 14$ True | $7 > 14$ False

$m = 21-14 = 7$ | $n = 14-7 = 7$

o/p: 7

The program computes

(~~A~~) x ÷ y using repeated subtraction

(~~B~~) x mod y using repeated subtraction

(C) the greatest common divisor of x and y

(~~D~~) the least common multiple of x and y

#Q. The for loop

```
for (i=0; i<10; ++i)
printf("%d", i&1);
```

prints

A. 0101010101

B. 0111111111

C. 0000000000

D. 1111111111

$1 = (0001)$

$x = (---*)$ → $1 = 1$ (odd numbers), $0 = 0$ (Even numbers)

$\& = 000* \Rightarrow 1/0$

$i =$	0	1	2	3	4	5	6	7	8	9
$i\&1 =$	0	1	0	1	0	1	0	1	0	1

#Q. What will be the output of the following C code?

```
#include <stdio.h>
main() {
    int i;
    for(i=0;i<5;i++) {
        int i=10;
        printf("%d" ,i);
        i++;
    }
    return 0;
}
```

Two different 'i' values

outer i	inner i
i=0	i=10, Printed 10 i=11
i=1	i=10, Printed 10 i=11
i=2	i=10, Printed 10 i=11
i=3	" "
i=4	" "

o/p: 10 10 10 10 10

A. 10 11 12 13 14

B. 10 10 10 10 10 ✓

C. 0 1 2 3 4

D. Compilation error

#Q. What is the output of the following C program?

```c
#include<stdio.h>
void main(void){
    int shifty;
    shifty=0570;
    shifty=shifty>>4;
    shifty=shifty<<6;
    printf("The value  of shifty is %o \n",shifty);
}
```

octal number

A. The value of shifty is 15c0
B. The value of shifty is 4300
C. The value of shifty is 5700
D. The value of shifty is 2700

0 5 7 0

Binary

000 101 111 000

>>4

000000010111

<<6

010111000000

Octal

2 7 0 0

Topic : Practice Session - 1

#Q. Consider the following program fragment

```
i=6720; j=4;
while (i%j)==0
{
   i=i/j;
   j=j+1;
}
```

On termination j will have the value

A. 4
B. 8
C. 9
D. 6720

6720 % 4 == 0 True
i = 1680
j = 5

1680 % 5 == 0 True
i = 336
j = 6

336 % 6 == 0 True
i = 56
j = 7

56 % 7 == 0 True
i = 8
j = 8

8 % 8 == 0 True
i = 1
$\boxed{j = 9}$

1 % 9 == 0 False

H/W

#Q. How many lines of output does the following C code produce?

```
float i=2.0;
float j=1.0;
float sum = 0.0;
main(){
    while (i/j > 0.001)   {
        j+=j;
        sum=sum+(i/j);
        printf("%f\n", sum);
    }
}
```

A. 8 C. 9

B. 10 D. 11

Topic : Practice Session - 1

H/W

#Q. Consider the following C program, It produces ________

```
main()
{
float sum= 0.0, j=1.0,i=2.0;
while(i/j>0.001){
   j=j+1;
   sum=sum+i/j;
   printf("%f/n", sum);
 }
}
```

A. 0 – 9 lines of output
B. 10 – 19 lines out output
C. 20 – 29 lines of output
D. More than 29 lines of output

H/W

#Q. What does the following program do when the input is unsigned 16 bit integer?

```
#include<stdio.h>
main(){
 unsigned int num;
 int i;
 scanf("%u", &num);
 for(i=0;i<16;i++){
  printf("%d", (num<<i&1<<15)?1:0);
 }
}
```

A. It prints all even bits from num
B. It prints all odd bits from num
C. It prints binary equivalent of num
D. None of above

H/W

#Q. Consider the following C function definition

```
int f (int x, int y) {
for (int i = 0; i<y; i++) {
x=x+x+y;
}
return x;
}
```

Which of the following statements is/are TRUE about the above function?

(A) If the inputs are x= 20, y =10, then the return value is greater than 2^{20}

(B) If the inputs are x = 20, y = 20, then the return value is greater than 2^{20}

(C) If the inputs are x = 20, y = 10, then the return value is less than 2^{10}

(D) If the inputs are x = 10, y = 20, then the return value is greater than 2^{20}

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THANK - YOU